

Assignment: _01

Submission Data: 19-12-25

Real-World Scenario

MegaGrid Corporation is building a next-generation intelligent transportation and energy routing system called UniFlow Network (UFN). This system integrates urban traffic networks, electric-vehicle charging infrastructures, and smart power-distribution grids into a single heterogeneous, weighted, multi-layer graph.

The graph contains:

- Road layer: Nodes represent intersections; edges represent roads with weights based on real-time congestion, accident probability, weather risk, and energy consumption.
- Charging layer: Nodes represent charging stations; edges represent charging slot availability, queue time, charging cost, and renewable energy contribution.
- Energy-distribution layer: Substations connected through transmission lines, weighted by load variance, failure probability, and transmission loss.

The goal is to compute multi-criteria shortest paths for different stakeholders such as self-driving taxis, emergency drones, and power grid optimizers. The environment is dynamic, requiring efficient re-optimization.

Task 1 – Graph Modeling (2 Marks)

Construct a multi-layer weighted graph (100+ nodes). Define node and edge types, weight formulas, and justify whether the graph is directed, undirected, or hybrid.

Task 2 – Implement Three Shortest Path Algorithms (4 Marks)

Implement from scratch:

1. Dijkstra's Algorithm
2. Bellman-Ford Algorithm
3. A* (A-Star) Algorithm

Include pseudocode, implementation, and complexity analysis. Design a domain-specific heuristic for A* and explain admissibility.

Task 3 – Multi-Criteria Path Optimization (2 Marks)

Compute shortest paths for:

- Taxi fleet (min time)
- Drone unit (min risk)
- Power grid optimizer (min energy loss)

Then design a multi-objective model using weighted sum, Pareto frontier, or lexicographic optimization.

Task 4 – Dynamic Graph Re-Optimization (2 Marks)

Simulate real-time changes in weights. Implement incremental re-optimization and compare with full recalculation. Provide charts for 10–20 dynamic updates.